Constitutional Reframing

Reorganizing Domains Around Present Realizability Rather Than Inherited Constraint

The old forms were not wrong. They were answers to vanished questions.

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Abstract

Every domain inherits structures from the conditions under which it first became workable. Institutions, technologies, crafts, scientific disciplines, organizational forms, and conceptual categories emerge as solutions to specific constraints. Over time, those solutions often become naturalized. What began as adaptation becomes ontology. What was once contingent becomes treated as essential.

This paper proposes **Constitutional Reframing**, a general analytical framework for distinguishing between a domain's enduring requirements and the historical constraints that originally shaped its architecture. The central claim is that many contemporary systems remain organized around limitations that no longer exist. As technological, informational, and organizational capabilities evolve, the inherited constitutional structure of a domain may cease to represent the most coherent expression of what is presently realizable.

Constitutional Reframing does not seek to discard tradition. Rather, it seeks to separate necessity from inheritance. The framework asks a simple but powerful question:

If the domain were invented today, with today's capabilities, would we organize it this way?

The paper develops a three-layer epistemic structure. Layer 1 identifies historically inherited constraints. Layer 2 evaluates whether contemporary capabilities dissolve those constraints. Layer 3 examines whether the conceptual vocabulary of the domain itself should be reorganized around present realizability rather than historical workaround.

Examples from propulsion systems, event-driven craft processes, information systems, and institutional design illustrate how domains often retain architectures optimized for vanished conditions. The paper concludes that many contemporary systems may be better understood not as fixed structures but as evolving

constitutional arrangements whose legitimacy depends upon current capability rather than historical precedent.

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1. Introduction

Human beings inherit more than traditions.

We inherit solutions.

Most domains do not begin as expressions of ideal design. They begin as accommodations to reality. Scarcity, distance, ignorance, material limits, communication delays, computational weakness, and measurement constraints shape what becomes possible.

The resulting structures are often brilliant.

Yet brilliance under one set of constraints does not guarantee optimality under another.

A sailing ship is not an error because engines exist.

A telegraph is not irrational because fiber optics exist.

A guild is not foolish because digital coordination exists.

The question is not whether historical structures were justified.

The question is whether the conditions that justified them still obtain.

Constitutional Reframing begins from this distinction.

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2. The Principle of Historical Inheritance

Every mature domain contains two layers:

1. The problem the domain addresses.
2. The architecture through which the problem was historically addressed.

These layers frequently become conflated.

The architecture becomes mistaken for the problem itself.

This confusion produces institutional inertia.

The domain forgets why its structures exist.

Eventually the structures are defended not because they solve a problem, but because they are familiar.

The original constraint disappears.

The solution remains.

Constitutional Reframing seeks to reverse this process.

Its purpose is not destruction.

Its purpose is remembrance.

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3. Vanished Questions

The subtitle of this paper contains its central thesis:

> The old forms were not wrong. They were answers to vanished questions.

Many inherited structures are artifacts of questions no longer asked.

Examples include:

- Mechanical transmissions solving variable-load power delivery before modern power electronics.
- Event-centered craft traditions solving observability problems before instrumentation.
- Hierarchical bureaucracies solving coordination problems before digital communication.
- Physical archives solving information persistence before distributed storage.
- Geographic centralization solving access problems before networked connectivity.

These structures often persist long after the questions disappear.

The answer survives.

The question fades.

Constitutional Reframing attempts to recover the question.

Only then can the answer be evaluated properly.

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4. Constitutional Structures

A constitutional structure is the set of assumptions that define what a domain considers possible. These assumptions are rarely explicit; they operate beneath methodology, beneath tools, and beneath language. They determine:

- What counts as normal.
- What counts as practical.
- What counts as realistic.
- What counts as impossible.

The most powerful constitutional assumptions are often invisible precisely because they are shared by everyone operating within the domain.

Constitutional Reframing makes them visible.

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5. The Realizability Principle

The framework rests on a simple proposition:

Domains should be organized around present realizability rather than inherited constraint.

Present realizability refers to what can actually be achieved using currently available capabilities.

These capabilities may include:

- Computational capacity.
- Instrumentation.
- Materials science.
- Energy systems.
- Information networks.
- Automation.
- Distributed coordination.
- Machine intelligence.
- Advanced sensing.

A domain becomes constitutionally outdated when its structure reflects limitations that these capabilities have already dissolved.

The question is not whether the old structure still functions.

The question is whether it remains the most coherent expression of what is possible.

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6. The Three-Layer Framework

Layer 1: Historical Constraint Analysis

What conditions originally produced the architecture?

What limitations were being solved?

What alternatives were unavailable?

Layer 1 is descriptive.

Its purpose is genealogy.

It asks how the domain became what it is.

Layer 2: Capability Reassessment

Do those constraints still exist?

Have new technologies, methods, or forms of coordination changed what is feasible?

Layer 2 is empirical.

Its purpose is evaluation.

It asks whether the inherited architecture remains necessary.

Layer 3: Constitutional Reorganization

If the domain were designed today, from first principles, would it retain the same categories?

Would it preserve the same boundaries?

Would it organize itself around the same assumptions?

Layer 3 is normative.

Its purpose is reframing.

It asks what the domain should become when reorganized around current possibility.

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7. Propulsion as a Constitutional Example

Traditional propulsion architectures emerged under severe electromechanical limitations.

Engines produced torque.

Transmissions adapted torque.

Wheels received torque.

This arrangement became synonymous with propulsion itself.

Yet modern power electronics, high-bandwidth motor control, advanced energy storage, and software-defined governance substantially alter what is realizable.

The constitutional question becomes:

Is propulsion fundamentally the management of gears and mechanical coupling?

Or is it fundamentally the governance of energy flows?

When viewed through contemporary capability, propulsion begins to appear less like a mechanical hierarchy and more like a coordinated energy system.

The historical architecture solved a problem.

The problem changed.

The architecture remained.

This is a constitutional phenomenon.

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8. Craft as a Constitutional Example

Traditional crafts often organize themselves around observable events.

First crack.

Oven spring.

Gelation.

Cone collapse.

These events became central because practitioners lacked direct access to latent process states.

The events solved an observability problem.

Modern instrumentation changes that condition.

The constitutional question becomes:

Is the event the process?

Or is the event merely evidence of the process?

When viewed through state estimation and control theory, many craft traditions appear to have elevated observables into organizing principles because observables were historically scarce.

Again, the structure was reasonable.

The conditions changed.

The structure remained.

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9. Scientific Publishing as a Constitutional Example

A third domain illustrates the framework's applicability beyond physical infrastructure and sensory craft.

Scientific publishing emerged in the seventeenth century with the founding of the *Philosophical Transactions of the Royal Society* (1665) and *Le Journal des Scavans*. The architecture that became dominant had four interlocking elements: discrete *journals* that bundled papers for periodic distribution; *peer review* in which a small number of editor-selected reviewers assessed manuscripts for a future evidence-qualified revision; *citation* as the mechanism for tracing intellectual lineage; and *indexing* through institutional libraries that aggregated journals geographically.

Each of these elements solved a specific historical constraint. Bundled journals solved the high fixed cost of physical distribution: aggregating papers reduced per-paper printing and shipping cost. Pre-publication peer review solved the limited capacity of small intellectual networks to evaluate work post-hoc: errors and weak contributions had to be filtered for a future evidence-qualified revision because corrective communication afterward was slow and limited in reach. Citation tracking through bibliographic apparatus solved the difficulty of locating prior work in geographically distributed libraries. Institutional indexing solved the absence of any other mechanism for organizing the cumulative scientific corpus.

These constraints have substantially relaxed. Digital distribution has reduced per-paper distribution cost to near zero. Computational citation analysis can trace lineage across the entire literature instantly. Open repositories provide access independent of geographic or institutional affiliation. Post-publication review (preprint servers with comment threads, replication studies as standalone papers, retraction databases, altmetric tracking) can correct or extend the literature at a cadence and scale unavailable to pre-publication review alone. Algorithmic discovery and recommendation have changed how researchers find relevant work.

Under the framework's central question — *if the domain were invented today, with today's capabilities, would we organize it this way?* — the answer for scientific publishing is plausibly no. The constitutional question becomes: is scientific publishing fundamentally the activity of bundling papers into branded journals with pre-publication gatekeeping? Or is it fundamentally the activity of recording, evaluating, and connecting knowledge claims across a community of researchers?

When viewed through contemporary capability, scientific publishing begins to appear less like a system of physical journals with attached evaluation processes and more like a coordinated knowledge-claim network with multiple evaluation, correction, and discovery mechanisms operating at different timescales. The journal as a discrete bundled artifact begins to look like a historical workaround for distribution costs that no longer exist. Pre-publication peer review begins to look like one of several possible evaluation mechanisms rather than the constitutive evaluation mechanism. The branded journal hierarchy begins to look like a status-allocation system inherited from when journal access was the primary indicator of work quality.

This is a constitutional analysis, not an argument that journals should be abolished or that peer review is worthless. The framework's claim is more modest: the *organization* of scientific publishing around branded journals with pre-publication gatekeeping reflects historical constraints that have substantially relaxed, and the more accurate post-digital description treats publication as one event in a longer cycle of recording, distributing, evaluating, correcting, and connecting knowledge claims. The inherited architecture remains operationally viable; whether it remains the most coherent expression of what is now possible is a Layer 3 question whose answer has substantial implications for science policy, research evaluation, and the structure of academic careers.

The same constitutional phenomenon appears.

The historical architecture solved a problem.

The capabilities that defined the problem have changed.

The architecture has not.

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10. The Distinction Between Essence and Workaround

A core objective of Constitutional Reframing is separating essence from workaround.

Essence refers to the enduring problem a domain addresses.

Workaround refers to the historically contingent method used to address it.

The two are often confused.

For example:

- **Domain:** Transportation
  Essence: Movement
  Historical Workaround: Mechanical transmission
- **Domain:** Communication
  Essence: Information exchange
  Historical Workaround: Physical mail
- **Domain:** Knowledge storage
  Essence: Persistence
  Historical Workaround: Paper archives
- **Domain:** Craft control
  Essence: State estimation
  Historical Workaround: Event observables
- **Domain:** Coordination
  Essence: Collective action
  Historical Workaround: Bureaucratic hierarchy

When workaround becomes mistaken for essence, innovation appears threatening.

When essence is distinguished from workaround, innovation becomes reinterpretation rather than disruption.

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11. Constitutional Lag

A domain experiences constitutional lag when its conceptual structure evolves more slowly than its practical capabilities. Constitutional lag manifests as persistent complexity, redundant intermediaries, artificial bottlenecks, legacy categories, explanatory confusion, and resistance to reorganization. The greater the gap between capability and constitution, the greater the lag — and many modern institutions exhibit precisely this condition: their technologies have changed; their constitutions have not.

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12. Reframing Versus Revolution

Constitutional Reframing should not be confused with revolution. Revolution seeks replacement; reframing seeks reinterpretation. The difference is crucial. A constitutional reframe often preserves most of the physical system while transforming its conceptual organization — the components remain, the meaning changes, and the structure becomes intelligible in a new way. The goal is coherence, not destruction.

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13. A Method for Constitutional Analysis

The framework proposes five diagnostic questions:

Question 1

What problem was the current architecture originally solving?

Question 2 Which constraints made that architecture necessary?

Question 3 Which of those constraints still exist?

Question 4 Which contemporary capabilities change the answer?

Question 5 If beginning today, what architecture would emerge?

Together these questions constitute a practical method for constitutional investigation.

14. Risks and Failure Modes

Not every inherited structure is obsolete.

Some survive because they embody genuine necessities.

Constitutional Reframing therefore carries risks:

- Mistaking maturity for inertia.
- Mistaking complexity for inefficiency.
- Underestimating hidden functions.
- Overestimating technological capability.
- Confusing novelty with improvement.

The framework requires skepticism toward both tradition and innovation. Its purpose is neither preservation nor disruption; its purpose is discrimination.

When the Framework Does Not Apply

Beyond misapplication risks, the framework has genuine limits of applicability. Constitutional Reframing is unlikely to produce useful reorganization in domains where any of the following conditions obtain.

The inherited structure tracks a constraint that has not actually relaxed. Some structures appear to be historical workarounds but are actually solving problems that current capabilities do not address. The framework's diagnostic depends on identifying which historical constraints have *substantially* relaxed; misidentifying the constraint structure produces reframings that look formally elegant but fail when implemented. Constitutional Reframing requires accurate genealogy, not just intuitive plausibility.

The reframing produces conceptual reorganization without tracking practical improvement. A reframing may be internally coherent and may correctly identify a historical workaround, yet produce a reorganized structure that delivers no operational advantage over the inherited one. The Layer 3 normative claim must connect to either Layer 1 (architectural realizability under present capability) or Layer 2 (comparative advantage along named metrics); reframings that float free of both lower layers are intellectual exercises rather than actionable analyses.

The inherited structure has acquired functions beyond its original purpose. A workaround that solved a vanished constraint may, over time, have come to perform additional functions whose value is independent of the original constraint. Pre-publication peer review may have originated as an editorial filtering mechanism but may now also serve as a community-formation, status-allocation, and methodological-training mechanism whose value persists even if the original filtering function is obsolete. Reframings that attend only to the original constraint may underestimate these accreted functions.

The domain's apparent inheritance is actually convergent reinvention. Some structures look like inheritance from earlier eras but are actually the equilibrium that present capabilities themselves produce. The framework requires distinguishing between *path-dependent inheritance* (structures that persist because of historical contingency) and *convergent equilibria* (structures that present conditions would produce regardless of history). Misidentifying the latter as the former produces reframings that propose returning to structures the domain already would have arrived at independently.

The reframing operation requires deep domain familiarity that the analyst lacks. Constitutional Reframing in any specific domain depends on accurate recognition of which structures are constitutive and which are workaround — a judgment that requires sustained engagement with the domain's working practitioners and accumulated empirical experience. Framework instances developed without that familiarity produce reframings that may be formally correct but operationally naive.

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15. Constitutional Convergence

As domains are reorganized around present capability, unexpected structural similarities emerge.

Propulsion begins to resemble energy governance.

Craft begins to resemble state estimation.

Organizations begin to resemble information systems.

Infrastructure begins to resemble distributed computation.

These similarities do not imply sameness.

They imply shared constitutional patterns.

Domains that once appeared unrelated reveal common underlying structures once historical constraints are removed from analysis.

This convergence may be one of the most significant consequences of constitutional reframing.

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16. Constitutional Computing

A broader research program may follow, though the present paper does not establish it.

The Constitutional Reframing analytical operation, repeated across many domains, may eventually constitute an empirical basis for a more general study: how organizations of any kind — technical systems, institutions, communities,

conceptual frameworks — navigate changing possibility spaces under shifting constraints. If a sufficient number of domain-specific reframings reveal common structural patterns (the constitutional convergence noted in Section 15), the cumulative pattern may admit formal treatment in its own right.

The conjecture is that constitutional organization is structurally similar to control under viability constraints (Aubin, 1991): the task shifts from executing fixed procedures to maintaining viable trajectories through a changing realizability space. The relevant question becomes: *what forms of organization best navigate a changing space of realizable possibilities?* This question has analogues in viability theory, in evolutionary economics, in adaptive systems theory, and in the philosophy of socio-technical evolution; whether these analogues compose into a single coherent framework is itself a research question.

The phrase **Constitutional Computing** is offered as an aspirational label for this larger program. The present paper does not establish that the program is well-formed, nor that the analogues are deep rather than superficial; it offers Constitutional Reframing as one operation that may belong to that broader study and notes that other operations (constitutional inheritance analysis, capability mapping, viability assessment under shifting constraints) may be required to fill out the program. The label is therefore a placeholder for future work rather than a claim of achievement, and the reader should treat this section as program statement rather than result.

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17. The Framework Examines Itself

Before closing, the framework should be subjected to the same examination it proposes for other inherited structures. The discipline articulated here is itself a constitutional structure, and intellectual honesty requires acknowledging its own constitutional conditions before applying it elsewhere. This section performs that self-audit; the closing paragraphs that follow return to the central claim.

The framework's constitutional moment. Constitutional Reframing as articulated here emerges from a specific historical moment whose features shape what the framework can see and how it operates. The framework presupposes a particular configuration of present capabilities — digital communication, computational analysis, abundant information access, distributed coordination — as the relevant background against which historical workarounds become visible as workarounds. A framework articulated under different background conditions would find different inherited structures salient. The framework therefore does not claim a privileged vantage outside historical conditions; it claims that *given the present configuration of capabilities*, certain inherited structures appear as workarounds whose continued use is worth examining. The framework's own analytic vocabulary may itself appear, from a future vantage, as the historically contingent response to particular contemporary constraints.

The framework's epistemic privileges. The framework presupposes certain capabilities on the part of the analyst: access to historical genealogies of the domain in question, familiarity with current technological and organizational possibilities, and the intellectual time and freedom to perform genealogical analysis. These analytic capabilities are themselves unevenly distributed; the framework therefore tends to be more accessible to analysts embedded in environments where historical-genealogical and contemporary-technological information are both readily available. This is a constraint on the framework's universality that should be acknowledged: Constitutional Reframing is itself a contemporary intellectual practice with its own constitutional structure, and may be less applicable in contexts where the relevant historical and technological information is not equally accessible.

The codification problem. The first two papers in this series performed Constitutional Reframing in specific domains (event-driven craft processes; propulsion architecture) without naming the operation. The present paper names and codifies the operation, with the intention of making it teachable and reproducible. Codification has costs as well as benefits. The original demonstrations emerged from sustained engagement with specific domains; the codified method risks being applied wherever the template seems to fit, including in domains where the analyst lacks the deep familiarity that made the original demonstrations correct rather than merely formally available. The framework as codified is therefore necessary but not sufficient: it provides vocabulary and structure for the analytical operation, but cannot substitute for the substantive engagement with a specific domain that any particular Constitutional Reframing requires.

The framework's own falsifiability. The framework's claims at Layer 3 (that Constitutional Reframing is a general analytical operation applicable across diverse domains) are themselves subject to the framework's own Layer 2 evaluation. The claim to generality currently rests on three worked examples: event-driven craft, propulsion architecture, and scientific publishing. Three instantiations are sufficient to demonstrate the operation can be applied across more than one domain but are insufficient to establish that the operation reveals genuine cross-domain patterns rather than convenient analogies. The framework's claim to generality is therefore a hypothesis that requires additional domain-specific work to test. Should reframings in other domains repeatedly fail to produce useful reorganizations, or should they succeed only by stretching the framework's vocabulary in ways that void its specificity, the generality claim would be appropriately weakened.

The framework, examined by itself, is a constitutional structure of recent emergence, dependent on present analytical capabilities, codified at a cost to its original demonstrative immediacy, and supported by a small number of worked examples whose extension to broader domains remains future work. With that acknowledgment in place, the central claim returns.

Conclusion

Civilizations inherit answers. Rarely do they inherit the questions.

The old forms were not wrong. They were answers to vanished questions. To see them as they were is to recover what they once solved — and to know, by that recovery, whether the conditions that made them necessary still hold.

Where they do, the inherited form is essence.

Where they do not, the inherited form is workaround. And a domain organized around workaround treats as fundamental what is merely historical.

Appendix A: Constitutional Reframing Template

For any domain:

1. Identify the inherited architecture.
2. Identify the historical constraints.
3. Map current capabilities.
4. Determine which constraints remain active.
5. Reconstruct the domain from present realizability.
6. Compare reconstructed architecture against inherited architecture.
7. Evaluate costs and benefits of transition.

Appendix B: Three-Layer Evidential Structure

- **Layer:** Layer 1
  Question: Why did the structure emerge?
  Evidence Type: Historical
- **Layer:** Layer 2
  Question: Do the constraints still exist?
  Evidence Type: Empirical
- **Layer:** Layer 3
  Question: Should the domain be reorganized?
  Evidence Type: Normative

Appendix C: Constitutional Vocabulary

Constitutional Structure The deep organizational assumptions of a domain.

Historical Inheritance A structure preserved from conditions that produced it.

Constitutional Lag The gap between current capability and inherited organization.

Present Realizability The set of outcomes achievable using contemporary capability.

Vanished Question A historical problem whose solution persists after the problem itself disappears.

Constitutional Reframing The process of reorganizing a domain around present realizability rather than inherited constraint.

Corpus-Wide Reference Positioning

This paper argues that every domain inherits structures from the conditions under which it first became workable. Its nearest external comparison fields are governance, software, formal. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. Modularity, secure-development practice, and reproducibility reduce classes of failure but do not guarantee architectural coherence or user sovereignty. Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action.* Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E2**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

[**E3**] Lawrence Lessig. *Code: Version 2.0.* Basic Books, 2006. https://codev2.cc/ **Relationship:** Predecessor argument that software architecture regulates behavior; the paper's contribution may be to make such regulation inspectable and revisable.

[**E4**] OECD. *Recommendation of the Council on Artificial Intelligence.* OECD/LEGAL/0449, adopted 2019 and updated 2024. legalinstruments.oecd.org source **Relationship:** Adjacent policy standard for accountability, traceability, transparency, safety, and human-centered values.

[**E5**] David L. Parnas. "On the Criteria To Be Used in Decomposing Systems into Modules." *Communications of the ACM* 15(12), 1972, pp. 1053-1058. DOI: 10.1145/361598.361623. https://doi.org/10.1145/361598.361623 **Relationship:** Foundational modularity treatment used to compare information hiding with explicit semantic and governance interfaces.

[**E6**] Ward Cunningham. "The WyCash Portfolio Management System." *OOPSLA '92 Experience Report*, 1992. DOI: 10.1145/157709.157715. https://doi.org/10.1145/157709.157715 **Relationship:** Original technical-debt metaphor used to distinguish implementation debt from semantic debt.

[**E7**] Karen Scarfone, Murugiah Souppaya, and Donna Dodson. *Secure Software Development Framework (SSDF) Version 1.1.* NIST SP 800-218, 2022. DOI: 10.6028/NIST.SP.800-218. https://doi.org/10.6028/NIST.SP.800-218 **Relationship:** Adjacent secure-development framework for preparation, protection, production, and vulnerability response.

Ableman corpus

[**A1**] Adam Ableman. *Ableman Constitutional Governance*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-79353AF163A4B0F1;
source SHA-256
FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Theory*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-D2D97C4042FD281F;
source SHA-256
FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Governance Engineering*. Ableman Corpus current edition; work identity PAPER.PUB.18749870;
source SHA-256
372169F315D192733856B1F4A3F1038BBC73F57CF3ED9ACABAE72B965A69C8D0.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Universal Governance Kernel (UGK)*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-1F9BFB2BF8370226;
source SHA-256
AA5106EA71E5DF120E141435C280DFC532715CA5DE0C75B8862BB7784ECCFACD.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Legitimacy Without Self-Ratification*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-EF2382F9397335AC;
source SHA-256
0186A39C808F8FDDB5C8F42384DF1A26678928434BD50A0EAE8D4947390F68A2.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Sovereign Software*. Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-24C6159B98023A9C;
source SHA-256
6AAFE79E438B980B75D1910A897F7BA716A10CF1187B68CB1F71405F36725A18.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Constitutional Software Analysis (CSoftA): A Governance-Based Classification Method for Software Systems.* Ableman Corpus current edition; work identity `PAPER.PUB.20472196`;
source SHA-256
`3670B28B7C1AF5467BE4002C47B5F964F8FAF6F257125C2F84CE06904E4FD0DE`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

The framework draws on three literatures: the formal apparatus of viability theory and control under constraint; the philosophy and history of technology, with particular attention to genealogical analyses of inherited structures; and adjacent literatures on path dependence, institutional inertia, and the social construction of technological systems. The companion papers ground the framework's first two worked examples.

Companion Papers

Ableman Adam Ableman (2025a). *Direct-Coupled Dynamic Torque Reactor: Programmable Inertia, Torque Vector Shaping, and Constitutional Electromechanical Governance in Post-Mechanical Propulsion.* Constitutional Computing Research Manuscript.

Ableman Adam Ableman (2025b). *Event-Driven Craft Processes as State-Estimation Problems: Observability, Viability Constraints, and Latent-State Control in Traditional Craft Practice.* Constitutional Computing Research Manuscript.

Viability Theory and Control under Constraint

Aubin, J.-P. (1991). *Viability Theory.* Birkhauser.

Aubin, J.-P., Bayen, A., & Saint-Pierre, P. (2011). *Viability Theory: New Directions* (2nd ed.). Springer.

Philosophy and History of Technology

Arthur, W. B. (2009). *The Nature of Technology: What It Is and How It Evolves.* Free Press. Treatment of technology as combinatorial recombination of existing components; relevant to the framework's account of how inherited architectures persist beyond their original conditions.

Borgmann, A. (1984). *Technology and the Character of Contemporary Life: A Philosophical Inquiry.* University of Chicago Press. The device paradigm and the distinction between focal practices and technological availability; relevant to the framework's distinction between essence and workaround.

Heidegger, M. (1977). The Question Concerning Technology. In *The Question Concerning Technology and Other Essays* (W. Lovitt, Trans.). Harper & Row. (Original work published 1954.) Foundational treatment of technology as a mode of revealing rather than as instrumentation; relevant to the framework's claim that inherited architectures shape what counts as possible within a domain.

Ihde, D. (1990). *Technology and the Lifeworld: From Garden to Earth*. Indiana University Press. Phenomenological analysis of how technologies mediate human experience of their domains; relevant to the framework's account of how constitutional structures become invisible to participants.

Verbeek, P.-P. (2005). *What Things Do: Philosophical Reflections on Technology, Agency, and Design*. Pennsylvania State University Press. Postphenomenological analysis of technology's mediating role.

Winner, L. (1986). *The Whale and the Reactor: A Search for Limits in an Age of High Technology*. University of Chicago Press. "Do Artifacts Have Politics?" essay particularly relevant to the framework's analysis of inherited structures as embedding historical assumptions.

Path Dependence and Institutional Inheritance

David, P. A. (1985). Clio and the economics of QWERTY. *American Economic Review*, 75(2), 332–337. Foundational treatment of how historically contingent choices become locked-in as standards; the QWERTY keyboard as paradigmatic example of structure outliving the constraint that produced it.

North, D. C. (1990). *Institutions, Institutional Change and Economic Performance*. Cambridge University Press. Path-dependent persistence of institutional structures.

Page, S. E. (2006). Path dependence. *Quarterly Journal of Political Science*, 1(1), 87–115. Formal treatment of path dependence across multiple disciplines.

Pierson, P. (2000). Increasing returns, path dependence, and the study of politics. *American Political Science Review*, 94(2), 251–267. Path dependence as analytical framework for political institutions.

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Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press. Foundational treatment of how scientific frameworks shift; relevant background for the framework's Layer 3 normative claims about conceptual reorganization.

Lakatos, I. (1970). Falsification and the methodology of scientific research programmes. In I. Lakatos & A. Musgrave (Eds.), *Criticism and the Growth of*

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Note on citation scope. The reference list above is representative rather than comprehensive. The framework draws on these literatures but does not claim to integrate them; full integration of Constitutional Reframing with path-dependence economics, philosophy of technology, and socio-technical systems theory remains future work. Domain-specific applications of the framework (such as the companion papers' work in propulsion and craft) require additional domain-specific citation.

Standing and Claim Boundary

This paper is a conceptual, formal, or architectural contribution unless it explicitly binds a statement to a cited external source or an exact carried evidence artifact. Unbound statements about implementation, validation, performance, safety, universality, or proof are research claims or design obligations, not established operational facts. Corpus-internal references establish conceptual lineage and comparison, not independent external confirmation.

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The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

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Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus

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DOI: 10.5281/zenodo.23848668